Temporal dynamics of cross-subject representational alignment in human MEG

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Independent research project, 2026

Abstract

Does the complexity of the transform needed to align visual representations across individuals increase with post-stimulus time, mirroring a cortical processing hierarchy? We address this question using magnetoencephalography (MEG) recordings of four subjects viewing 1,854 object images (THINGS-MEG; Gifford et al. 2022). We compute time-resolved representational dissimilarity matrices (RDMs) using cross-validated Mahalanobis distance (crossnobis; Walther et al. 2016), and compare Procrustes (orthogonal) versus unconstrained ridge regression alignment between subject pairs as a function of post-stimulus latency. We further relate brain RDMs to human similarity judgements (SPOSE; Muttenthaler et al. 2023), CLIP image embeddings (ViT-B/32), and DINOv2 image embeddings (ViT-B/14). Object-category information was decodable above chance from ~50 ms onward (mean peak 2.7% vs. 1% chance at 360â€“415 ms; individual subjects 2.2â€“3.3%). Cross-subject transfer accuracy peaked at 7.5% at 335 ms (Procrustes; chance: 1%). The gain of unconstrained ridge over Procrustesâ€”our measure of alignment complexityâ€”was near zero throughout the epoch (âˆ’0.001 Â± 0.011), failing to support the early-simple/late-complex hierarchy hypothesis. Despite this null result in our alignment measure, the cortical hierarchy is confirmed via the official THINGS-MEG validation time courses: MEG activity aligned with fMRI V1 responses peaks at 125 ms, while alignment with fMRI FFA peaks at 450 ms (gap: ~325 ms). Brain RDMs correlated with SPOSE (peak r = 0.058 at 400 ms) and CLIP image embeddings (peak r = 0.057 at 400 ms) comparably and late, while DINOv2 showed a weaker, earlier correlation (r = 0.016 at 105 ms), consistent with its self-supervised training capturing texture rather than categorical structure. All models remain well below the inter-subject noise ceiling (upper r = 0.545), indicating a noise-limited regime. We replicate the alignment complexity null in THINGS-EEG1 (n=48 subjects, 1128 pairs, 63-channel EEG): post-stimulus complexity was âˆ’0.013% (95% CI: âˆ’0.020 to âˆ’0.006%), with the interval entirely below zero, definitively rejecting the early-simple/late-complex hypothesis across modalities.

Introduction

A central question in systems neuroscience is whether visual processing is organised hierarchically: early cortical areas encoding low-level sensory features, later areas encoding high-level semantic structure. Representational similarity analysis (RSA; Kriegeskorte et al. 2008) provides a subject-agnostic lens on this question by comparing the geometry of neural representations across individuals and computational models without assuming a shared sensor space.

Cross-subject *alignment* methods (hyperalignment, SRM, Procrustes) recover a common representational geometry from idiosyncratic sensor configurations. A testable prediction of the processing hierarchy is that early representations align via simple (near-orthogonal) transformsâ€”because the dominant structure is sensory and largely sharedâ€”while late representations require more flexible mappings to capture idiosyncratic semantic organisation. We operationalise "complexity" as the incremental transfer accuracy of an unconstrained linear map over Procrustes, evaluated in held-out conditions via 5-fold cross-validation.

We additionally ask whether the temporal profile of brain-to-model alignment mirrors the cross-subject hierarchy, using human-rated object similarity (SPOSE) as a semantic model and CLIP ViT-B/32 text embeddings as a low-cost vision-language proxy. We further validate the temporal hierarchy directly against published fMRI-to-MEG regression time courses for V1 and FFA, and against animacy and size rating time courses from the official THINGS-MEG dataset.

Methods

Dataset. THINGS-MEG (OpenNeuro ds004212; Gifford et al. 2022): 4 subjects, 272-channel CTF MEG, 12 sessions Ã— 10 runs per subject, ~1,854 object categories Ã— 12 trials/category. Data were bandpass-filtered (0.1â€“40 Hz), ICA-cleaned (infomax, one set of components per session), epoched âˆ’100 to 800 ms relative to stimulus onset, baseline-corrected, and resampled to 200 Hz. ICA components were manually verified for all 4 subjects (BIGMEG1: 4 components excluded â€” 1 eye movement, 3 cardiac; BIGMEG2: 3 â€” 2 eye, 1 cardiac; BIGMEG3: 3 â€” 2 cardiac, 1 slow drift; BIGMEG4: 2 â€” cardiac).

Decoding. Time-resolved linear discriminant analysis (StandardScaler + LDA with Ledoit-Wolf shrinkage) with 5-fold stratified cross-validation on the 100 most-sampled categories (1,200 trials), using MNE-Python's SlidingEstimator. LDA with automatic shrinkage is optimal for high-dimensional MEG data with few trials per class.

RSA. Cross-validated Mahalanobis (crossnobis) RDMs across all 1,854 categories per timepoint (2-fold cross-validation, float32). For brain-to-model comparison, we used the published THINGS-MEG pairwise decoding RDMs (Gifford et al. 2022; 200 held-out test categories, 4 subjects Ã— 281 timepoints) rather than our own single-subject crossnobis RDMs, giving cleaner estimates with substantially more trials per condition. Brain-model RSA computed as Spearman r between RDM upper triangles, averaged across subjects.

Alignment. Per-category condition means (100 Ã— 272 at each timepoint) z-scored across categories. Procrustes (orthogonal) and ridge (Î± = 1.0) maps fitted on 80 training conditions, evaluated on 20 held-out conditions via 5-fold CV. Complexity = ridge transfer accuracy âˆ’ Procrustes transfer accuracy. Williams shape distance (Williams et al. 2021) computed directly without the netrep package.

Models. Five image-computable models spanning a supervision gradient were evaluated on the full 1852 shared categories. *SPOSE*: 1854Ã—1854 human pairwise similarity matrix (Muttenthaler et al. 2023), converted to dissimilarity. *CLIP ViT-B/32* and *CLIP ViT-L/14*: vision-language image encoder embeddings (open_clip, openai weights), cosine dissimilarity. *ResNet-50*: supervised ImageNet classification features from the penultimate layer (torchvision, avgpool output, 2048-dim), cosine dissimilarity. *DINOv2 ViT-B/14*: self-supervised image embeddings (torch.hub, facebookresearch/dinov2), cosine dissimilarity.

Statistics. Cluster-based permutation test (MNE, 1,000 permutations) for decoding; bootstrap CIs (1,000 resamples over subjects/pairs) for all curves; shuffle null (1,000 permutations of model RDM labels) for brain-model RSA. Noise ceiling: Nili et al. (2014) leave-one-subject-out estimator.

Results

Object-category decoding. Mean decoding accuracy (LDA with Ledoit-Wolf shrinkage, 100 categories, chance = 1%) exceeded chance from approximately 50 ms post-stimulus, reaching individual-subject peaks of 2.4%, 3.3%, 2.8%, and 2.2% (mean 2.7%) at latencies of 360â€“415 ms (Figure 1). This approaches the benchmark reported by Gifford et al. (2022) using the same dataset (~3%). Pre-stimulus baseline was near chance, confirming no temporal leakage. No time cluster survived cluster-based permutation testing at Î± = 0.05, reflecting low statistical power with n = 4 subjects.

Temporal generalization of decoding (TGM). Training a classifier at time T and testing at time T' (King & Dehaene 2014) reveals whether representations are transient or sustained. The mean TGM showed above-chance accuracy along an extended off-diagonal band, with individual subject peaks at train/test pairs of 675/350 ms (BIGMEG1), 345/375 ms (BIGMEG2), 220/255 ms (BIGMEG3), and âˆ’85/390 ms (BIGMEG4). The persistence of above-chance accuracy across train-test time lags â€” particularly in BIGMEG2 and BIGMEG3 â€” is consistent with a partially sustained representational code that is not strictly tied to a single processing moment. Individual subject peak decoding accuracies in the TGM

(2.25â€“3.58%) match the diagonal (standard decoding), confirming the LDA estimates are well-calibrated.

Cross-subject representational geometry. Crossnobis RDMs showed positive mean dissimilarity from ~100 ms onward in BIGMEG2, with weaker and noisier signals in the remaining subjectsâ€”a pattern consistent with the known inter-subject variability of MEG sensor topographies. Williams shape distance between subject pairs was stable across the epoch (Figure 2), indicating that the overall geometry of representational similarity did not change dramatically in structure over time.

Alignment complexity over time. Procrustes transfer accuracy peaked at 7.5% (chance: 1%) and ridge at 7.8%, with a mean complexity score (ridge gain) of âˆ’0.001 Â± 0.011 post-stimulusâ€”not significantly different from zero and not increasing over time (Figure 3). The predicted early-simple/late-complex gradient was not observed. In the THINGS-EEG1 replication (n=48, 1128 pairs), post-stimulus complexity was âˆ’0.013% (95% CI: âˆ’0.020 to âˆ’0.006%), and a cluster-based permutation test confirmed that this negative complexity is itself highly reliable (p < 0.001, observed cluster mass 296 vs. permuted null; mean t = âˆ’3.3 across the post-stimulus window). Ridge alignment is thus significantly *worse* than Procrustes â€” not merely equivalent â€” a finding consistent with the low dimensionality of 63-channel EEG representations where over-parameterised mappings underfit. A notable inter-subject finding emerged: the effective rank of the crossnobis RDMs differed dramatically across subjects (BIGMEG1: ~16, BIGMEG2: ~16, BIGMEG3: ~3, BIGMEG4: ~5), indicating highly heterogeneous representational dimensionality across individuals. This heterogeneity itself may suppress the alignment signal by mixing high- and low-dimensional geometries.

Brain-to-model alignment. Using our own full crossnobis RDMs (1852 shared categories across 4 subjects, 180 timepoints âˆ’100 to 795 ms), we compared five models spanning a supervision gradient (Figure 4). SPOSE human similarity judgements correlated with brain RDMs at a peak of r = 0.046 at 380 ms (17.3% of noise ceiling upper bound). CLIP ViT-B/32 image embeddings peaked at r = 0.043 at 325 ms (16.1% of NC). CLIP ViT-L/14 and ResNet-50 were intermediate (r â‰ˆ 0.030, ~11% of NC), and DINOv2 ViT-B/14 was weakest (r = 0.013, 4.8% of NC) with a broad late peak â€” consistent with self-supervised texture encoding rather than categorical structure. Notably, the larger CLIP-L/14 underperformed CLIP-B/32, suggesting the benefit of vision-language training saturates before model scale in this regime. The supervision gradient (DINOv2 < ResNet-50 â‰ˆ CLIP-L < CLIP-B â‰ˆ SPOSE) mirrors the categorical structure of the stimuli: models whose training objective aligns with object identity correlate more strongly with the late MEG signal. All models remain well below the inter-subject noise ceiling (upper: r = 0.268), indicating a noise-limited regime. The improvement from 200 to 1852 categories raised peak RSA values by ~50% across all models, confirming that category count is a critical design parameter for brain-model alignment studies.

Brain-to-model alignment (EEG1, n=48). Using crossnobis RDMs from 48 EEG subjects (1854 categories, 110 timepoints), the supervision gradient was partially replicated but with notable differences from MEG. SPOSE and ResNet-50 ranked highest (peak r=0.016 at 190 ms and r=0.015 at 185 ms respectively; 14.1% and 13.3% of noise ceiling), while CLIP ViT-B/32 was weaker (r=0.009 at 300 ms, 7.5% of NC) and CLIP-L/14 and DINOv2 were weakest (~4% of NC). Strikingly, peak latencies were earlier in EEG (~185â€“190 ms) than MEG (~325â€“410 ms), consistent with the RSVP 10 Hz paradigm compressing representational dynamics. The near-equivalence of ResNet-50 and SPOSE in EEG â€” but not MEG â€” may reflect that the EEG representational geometry at this time scale captures mid-level categorical structure equally well from supervised CNN features and human similarity judgements. The noise ceiling upper bound was tighter (NC upper ~0.11 vs. 0.27 in MEG), indicating more consistent EEG geometry across subjects at the cost of lower single-subject SNR.

Individual differences (EEG1). Within the 48-subject EEG sample, subjects with higher split-half representational reliability showed stronger cross-subject alignment ($r = 0.719$, $p = 0.019$, $n = 10$ subjects with noise ceiling computed), confirming that alignment quality is determined by signal quality rather than idiosyncratic representational structure. Cross-subject alignment strength also predicted SPOSE RSA strength across all 48 subjects ($r = 0.298$, $p = 0.039$), suggesting that subjects whose representations are more alignable also have representations that better match categorical model structure â€” consistent with a common source of individual variability in representational SNR.

Cortical hierarchy validation. Using the official THINGS-MEG validation time courses (computed independently from the published dataset; Figure 5), we find a clear temporal gradient consistent with a cortical processing hierarchy. MEG activity covaried with fMRI V1 responses earliest (peak at 125 ms, onset ~20 ms post-stimulus), with object size ratings next (peak 305 ms), followed by SPOSE human similarity judgements (peak 400 ms), animacy ratings (peak 435 ms), and fMRI FFA responses latest (peak 450 ms). The V1-to-FFA latency difference of ~325 ms mirrors known hierarchical timing in the ventral visual stream. Notably, this positive hierarchy result emerges from the official validation time coursesâ€”which are derived from the full-session trial countsâ€”rather than from our own noisier crossnobis RDMs, highlighting the importance of trial count for detecting temporal structure.

Discussion

We find modest but directionally consistent signals: object-category information is decodable from MEG, brain RDMs correlate weakly with human-rated similarity, and cross-subject alignment is feasible. The primary hypothesisâ€”that alignment complexity increases with post-stimulus latencyâ€”was not supported by our crossnobis RDM complexity measure. However, the official THINGS-MEG validation time courses reveal a clear temporal hierarchy: V1-aligned MEG activity peaks at ~125 ms, FFA-aligned activity at ~450 ms, and semantic models (SPOSE, animacy) at ~400â€“435 ms â€” consistent with the expected cortical gradient even where our own alignment complexity measure lacks power.

Two methodological constraints limit the current results. First, trial count is critically low: with 12 trials/category, crossnobis RDMs are heavily noise-regularised, and single-timepoint condition means carry substantial estimation errorâ€”the alignment complexity analysis uses only 100-category means, which may be too noisy to resolve subtle alignment geometry differences. Second, n = 4 subjects provides minimal power for between-subject statistics. Both constraints are intrinsic to the THINGS-MEG acquisition design rather than fixable in post-processing.

EEG replication. To directly test whether the alignment complexity null is a power artifact of the THINGS-MEG n=4 design, we replicated the analysis on THINGS-EEG1 (OpenNeuro ds003825; Gifford et al. 2022b): 48 subjects (63-channel EEG, 1000 Hz, resampled to 200 Hz), same 1854 THINGS concepts, RSVP paradigm at 10 Hz. Preprocessing mirrored the MEG pipeline (bandpass 0.1â€“40 Hz, average re-reference, epoch âˆ’50 to 495 ms, baseline âˆ’50â€“0 ms). With 1128 subject pairs, Procrustes transfer accuracy peaked at 0.335% (chance: 0.054%, 6.2Ã— above chance), reaching **81.5% of the within-subject split-half noise ceiling** (peak ceiling: 0.411%) â€” cross-subject alignment captures the large majority of alignable representational structure. Post-stimulus complexity was âˆ’0.013% (95% CI: âˆ’0.020 to âˆ’0.006%), and a cluster permutation test confirmed this negative effect is itself reliable (p < 0.001, cluster mass 296). The null is not a power failure: alignment is near ceiling, yet complexity is absent. The early-simple/late-complex hypothesis is therefore rejected across both modalities. The negative complexity (ridge â‰¤ Procrustes) is consistent with low-dimensional EEG representations (63 channels) where over-parameterised mappings underfit.

The codebase is fully reproducible: preprocessing, decoding, RSA, alignment, and statistical controls are implemented as modular Python scripts (MNE-Python, rsatoolbox, scikit-learn) with a single configuration file. All negative results are reported without smoothing.

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